

Chapter 21

1. **THINK** After the transfer, the charges on the two spheres are $Q - q$ and q .

EXPRESS The magnitude of the electrostatic force between two charges q_1 and q_2 separated by a distance r is given by the Coulomb's law (see Eq. 21-1):

$$F = k \frac{q_1 q_2}{r^2},$$

where $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. In our case, $q_1 = Q - q$ and $q_2 = q$, so the magnitude of the force of either of the charges on the other is

$$F = \frac{1}{4\pi\epsilon_0} \frac{q(Q - q)}{r^2}.$$

We want the value of q that maximizes the function $f(q) = q(Q - q)$.

ANALYZE Setting the derivative df/dq equal to zero leads to $Q - 2q = 0$, or $q = Q/2$. Thus, $q/Q = 0.500$.

LEARN The force between the two spheres is a maximum when charges are distributed evenly between them.

2. The fact that the spheres are identical allows us to conclude that when two spheres are in contact, they share equal charge. Therefore, when a charged sphere (q) touches an uncharged one, they will (fairly quickly) each attain half that charge ($q/2$). We start with spheres 1 and 2, each having charge q and experiencing a mutual repulsive force $F = kq^2/r^2$. When the neutral sphere 3 touches sphere 1, sphere 1's charge decreases to $q/2$. Then sphere 3 (now carrying charge $q/2$) is brought into contact with sphere 2; a total amount of $q/2 + q$ becomes shared equally between them. Therefore, the charge of sphere 3 is $3q/4$ in the final situation. The repulsive force between spheres 1 and 2 is finally

$$F' = k \frac{(q/2)(3q/4)}{r^2} = \frac{3}{8} k \frac{q^2}{r^2} = \frac{3}{8} F \Rightarrow \frac{F'}{F} = \frac{3}{8} = 0.375.$$

3. **THINK** The magnitude of the electrostatic force between two charges q_1 and q_2 separated by a distance r is given by Coulomb's law.

EXPRESS Equation 21-1 gives Coulomb's law, $F = k \frac{|q_1||q_2|}{r^2}$, which can be used to solve for the distance:

$$r = \sqrt{\frac{k|q_1||q_2|}{F}}.$$

ANALYZE With $F = 5.70 \text{ N}$, $q_1 = 2.60 \times 10^{-6} \text{ C}$ and $q_2 = -47.0 \times 10^{-6} \text{ C}$, the distance between the two charges is

$$r = \sqrt{\frac{k|q_1||q_2|}{F}} = \sqrt{\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(2.60 \times 10^{-6} \text{ C})(47.0 \times 10^{-6} \text{ C})}{5.70 \text{ N}}} = 1.39 \text{ m}.$$

LEARN The electrostatic force between two charges falls as $1/r^2$. The same inverse-square nature is also seen in the gravitational force between two masses.

4. The unit ampere is discussed in Section 21-4. Using i for current, the charge transferred is

$$q = it = (2.5 \times 10^4 \text{ A})(20 \times 10^{-6} \text{ s}) = 0.50 \text{ C}.$$

5. The magnitude of the mutual force of attraction at $r = 0.120 \text{ m}$ is

$$F = k \frac{|q_1||q_2|}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(3.00 \times 10^{-6} \text{ C})(1.50 \times 10^{-6} \text{ C})}{(0.120 \text{ m})^2} = 2.81 \text{ N}.$$

6. (a) With a understood to mean the magnitude of acceleration, Newton's second and third laws lead to

$$m_2 a_2 = m_1 a_1 \Rightarrow m_2 = \frac{(6.3 \times 10^{-7} \text{ kg})(7.0 \text{ m/s}^2)}{9.0 \text{ m/s}^2} = 4.9 \times 10^{-7} \text{ kg}.$$

(b) The magnitude of the (only) force on particle 1 is

$$F = m_1 a_1 = k \frac{|q_1||q_2|}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{|q|^2}{(0.0032 \text{ m})^2}.$$

Inserting the values for m_1 and a_1 (see part (a)) we obtain $|q| = 7.1 \times 10^{-11} \text{ C}$.

7. With rightward positive, the net force on q_3 is

$$F_3 = F_{13} + F_{23} = k \frac{q_1 q_3}{(L_{12} + L_{23})^2} + k \frac{q_2 q_3}{L_{23}^2}.$$

We note that each term exhibits the proper sign (positive for rightward, negative for leftward) for all possible signs of the charges. For example, the first term (the force exerted on q_3 by q_1) is negative if they are unlike charges, indicating that q_3 is being

pulled toward q_1 , and it is positive if they are like charges (so q_3 would be repelled from q_1). Setting the net force equal to zero $L_{23} = L_{12}$ and canceling k , q_3 , and L_{12} leads to

$$\frac{q_1}{4.00} + q_2 = 0 \Rightarrow \frac{q_1}{q_2} = -4.00.$$

8. In experiment 1, sphere C first touches sphere A , and they divided up their total charge ($Q/2$ plus Q) equally between them. Thus, sphere A and sphere C each acquired charge $3Q/4$. Then, sphere C touches B and those spheres split up their total charge ($3Q/4$ plus $-Q/4$) so that B ends up with charge equal to $Q/4$. The force of repulsion between A and B is therefore

$$F_1 = k \frac{(3Q/4)(Q/4)}{d^2}$$

at the end of experiment 1. Now, in experiment 2, sphere C first touches B , which leaves each of them with charge $Q/8$. When C next touches A , sphere A is left with charge $9Q/16$. Consequently, the force of repulsion between A and B is

$$F_2 = k \frac{(9Q/16)(Q/8)}{d^2}$$

at the end of experiment 2. The ratio is

$$\frac{F_2}{F_1} = \frac{(9/16)(1/8)}{(3/4)(1/4)} = 0.375.$$

9. **THINK** Since opposite charges attract, the initial charge configurations must be of opposite signs. Similarly, since like charges repel, the final charge configurations must carry the same sign.

EXPRESS We assume that the spheres are far apart. Then the charge distribution on each of them is spherically symmetric and Coulomb's law can be used. Let q_1 and q_2 be the original charges. We choose the coordinate system so the force on q_2 is positive if it is repelled by q_1 . Then the force on q_2 is

$$F_a = -\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} = -k \frac{q_1 q_2}{r^2}$$

where $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$ and $r = 0.500 \text{ m}$. The negative sign indicates that the spheres attract each other. After the wire is connected, the spheres, being identical, acquire the same charge. Since charge is conserved, the total charge is the same as it was originally. This means the charge on each sphere is $(q_1 + q_2)/2$. The force is now repulsive and is given by

$$F_b = \frac{1}{4\pi\epsilon_0} \frac{\left(\frac{q_1+q_2}{2}\right)\left(\frac{q_1+q_2}{2}\right)}{r^2} = k \frac{(q_1+q_2)^2}{4r^2}.$$

We solve the two force equations simultaneously for q_1 and q_2 .

ANALYZE The first equation gives the product

$$q_1 q_2 = -\frac{r^2 F_a}{k} = -\frac{(0.500 \text{ m})^2 (0.108 \text{ N})}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2} = -3.00 \times 10^{-12} \text{ C}^2,$$

and the second gives the sum

$$q_1 + q_2 = 2r \sqrt{\frac{F_b}{k}} = 2(0.500 \text{ m}) \sqrt{\frac{0.0360 \text{ N}}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2}} = 2.00 \times 10^{-6} \text{ C}$$

where we have taken the positive root (which amounts to assuming $q_1 + q_2 \geq 0$). Thus, the product result provides the relation

$$q_2 = \frac{-(3.00 \times 10^{-12} \text{ C}^2)}{q_1}$$

which we substitute into the sum result, producing

$$q_1 - \frac{3.00 \times 10^{-12} \text{ C}^2}{q_1} = 2.00 \times 10^{-6} \text{ C}.$$

Multiplying by q_1 and rearranging, we obtain a quadratic equation

$$q_1^2 - (2.00 \times 10^{-6} \text{ C})q_1 - 3.00 \times 10^{-12} \text{ C}^2 = 0.$$

The solutions are

$$q_1 = \frac{2.00 \times 10^{-6} \text{ C} \pm \sqrt{(2.00 \times 10^{-6} \text{ C})^2 - 4(-3.00 \times 10^{-12} \text{ C}^2)}}{2}.$$

If the positive sign is used, $q_1 = 3.00 \times 10^{-6} \text{ C}$, and if the negative sign is used, $q_1 = -1.00 \times 10^{-6} \text{ C}$.

(a) Using $q_2 = (-3.00 \times 10^{-12})/q_1$ with $q_1 = 3.00 \times 10^{-6} \text{ C}$, we get $q_2 = -1.00 \times 10^{-6} \text{ C}$.

(b) If we instead work with the $q_1 = -1.00 \times 10^{-6} \text{ C}$ root, then we find $q_2 = 3.00 \times 10^{-6} \text{ C}$.

LEARN Note that since the spheres are identical, the solutions are essentially the same: one sphere originally had charge $-1.00 \times 10^{-6} \text{ C}$ and the other had charge $+3.00 \times 10^{-6} \text{ C}$. What happens if we had not made the assumption, above, that $q_1 + q_2 \geq 0$? If the signs of

the charges were reversed (so $q_1 + q_2 < 0$), then the forces remain the same, so a charge of $+1.00 \times 10^{-6}$ C on one sphere and a charge of -3.00×10^{-6} C on the other also satisfies the conditions of the problem.

10. For ease of presentation (of the computations below) we assume $Q > 0$ and $q < 0$ (although the final result does not depend on this particular choice).

(a) The x -component of the force experienced by $q_1 = Q$ is

$$F_{1x} = \frac{1}{4\pi\epsilon_0} \left(-\frac{(Q)(Q)}{(\sqrt{2}a)^2} \cos 45^\circ + \frac{(|q|)(Q)}{a^2} \right) = \frac{Q|q|}{4\pi\epsilon_0 a^2} \left(-\frac{Q/|q|}{2\sqrt{2}} + 1 \right)$$

which (upon requiring $F_{1x} = 0$) leads to $Q/|q| = 2\sqrt{2}$, or $Q/q = -2\sqrt{2} = -2.83$.

(b) The y -component of the net force on $q_2 = q$ is

$$F_{2y} = \frac{1}{4\pi\epsilon_0} \left(\frac{|q|^2}{(\sqrt{2}a)^2} \sin 45^\circ - \frac{(|q|)(Q)}{a^2} \right) = \frac{|q|^2}{4\pi\epsilon_0 a^2} \left(\frac{1}{2\sqrt{2}} - \frac{Q}{|q|} \right)$$

which (if we demand $F_{2y} = 0$) leads to $Q/q = -1/2\sqrt{2}$. The result is inconsistent with that obtained in part (a). Thus, we are unable to construct an equilibrium configuration with this geometry, where the only forces present are given by Eq. 21-1.

11. The force experienced by q_3 is

$$\vec{F}_3 = \vec{F}_{31} + \vec{F}_{32} + \vec{F}_{34} = \frac{1}{4\pi\epsilon_0} \left(-\frac{|q_3||q_1|}{a^2} \hat{j} + \frac{|q_3||q_2|}{(\sqrt{2}a)^2} (\cos 45^\circ \hat{i} + \sin 45^\circ \hat{j}) + \frac{|q_3||q_4|}{a^2} \hat{i} \right)$$

(a) Therefore, the x -component of the resultant force on q_3 is

$$F_{3x} = \frac{|q_3|}{4\pi\epsilon_0 a^2} \left(\frac{|q_2|}{2\sqrt{2}} + |q_4| \right) = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{2(1.0 \times 10^{-7} \text{ C})^2}{(0.050 \text{ m})^2} \left(\frac{1}{2\sqrt{2}} + 2 \right) = 0.17 \text{ N}.$$

(b) Similarly, the y -component of the net force on q_3 is

$$\begin{aligned} F_{3y} &= \frac{|q_3|}{4\pi\epsilon_0 a^2} \left(-|q_1| + \frac{|q_2|}{2\sqrt{2}} \right) = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{2(1.0 \times 10^{-7} \text{ C})^2}{(0.050 \text{ m})^2} \left(-1 + \frac{1}{2\sqrt{2}} \right) \\ &= -0.046 \text{ N}. \end{aligned}$$

12. (a) For the net force to be in the $+x$ direction, the y components of the individual forces must cancel. The angle of the force exerted by the $q_1 = 40 \mu\text{C}$ charge on $q_3 = 20 \mu\text{C}$ is 45° , and the angle of force exerted on q_3 by Q is at $-\theta$ where

$$\theta = \tan^{-1} \left(\frac{2.0 \text{ cm}}{3.0 \text{ cm}} \right) = 33.7^\circ.$$

Therefore, cancellation of y components requires

$$k \frac{q_1 q_3}{(0.02\sqrt{2} \text{ m})^2} \sin 45^\circ = k \frac{|Q| q_3}{\left(\sqrt{(0.030 \text{ m})^2 + (0.020 \text{ m})^2} \right)^2} \sin \theta$$

from which we obtain $|Q| = 83 \mu\text{C}$. Charge Q is “pulling” on q_3 , so (since $q_3 > 0$) we conclude $Q = -83 \mu\text{C}$.

(b) Now, we require that the x components cancel, and we note that in this case, the angle of force on q_3 exerted by Q is $+\theta$ (it is repulsive, and Q is positive-valued). Therefore,

$$k \frac{q_1 q_3}{(0.02\sqrt{2} \text{ m})^2} \cos 45^\circ = k \frac{Q q_3}{\left(\sqrt{(0.030 \text{ m})^2 + (0.020 \text{ m})^2} \right)^2} \cos \theta$$

from which we obtain $Q = 55.2 \mu\text{C} \approx 55 \mu\text{C}$.

13. (a) There is no equilibrium position for q_3 *between* the two fixed charges, because it is being pulled by one and pushed by the other (since q_1 and q_2 have different signs); in this region this means the two force arrows on q_3 are in the same direction and cannot cancel. It should also be clear that off-axis (with the axis defined as that which passes through the two fixed charges) there are no equilibrium positions. On the semi-infinite region of the axis that is nearest q_2 and furthest from q_1 an equilibrium position for q_3 cannot be found because $|q_1| < |q_2|$ and the magnitude of force exerted by q_2 is everywhere (in that region) stronger than that exerted by q_1 on q_3 . Thus, we must look in the semi-infinite region of the axis which is nearest q_1 and furthest from q_2 , where the net force on q_3 has magnitude

$$\left| k \frac{|q_1 q_3|}{L_0^2} - k \frac{|q_2 q_3|}{(L + L_0)^2} \right|$$

with $L = 10 \text{ cm}$ and L_0 is assumed to be *positive*. We set this equal to zero, as required by the problem, and cancel k and q_3 . Thus, we obtain

$$\frac{|q_1|}{L_0^2} - \frac{|q_2|}{(L+L_0)^2} = 0 \Rightarrow \left(\frac{L+L_0}{L_0} \right)^2 = \frac{|q_2|}{|q_1|} = \left| \frac{-3.0 \mu\text{C}}{+1.0 \mu\text{C}} \right| = 3.0$$

which yields (after taking the square root)

$$\frac{L+L_0}{L_0} = \sqrt{3} \Rightarrow L_0 = \frac{L}{\sqrt{3}-1} = \frac{10 \text{ cm}}{\sqrt{3}-1} \approx 14 \text{ cm}$$

for the distance between q_3 and q_1 . That is, q_3 should be placed at $x = -14 \text{ cm}$ along the x -axis.

(b) As stated above, $y = 0$.

14. (a) The individual force magnitudes (acting on Q) are, by Eq. 21-1,

$$\frac{1}{4\pi\epsilon_0} \frac{|q_1|Q}{(-a-a/2)^2} = \frac{1}{4\pi\epsilon_0} \frac{|q_2|Q}{(a-a/2)^2}$$

which leads to $|q_1| = 9.0 |q_2|$. Since Q is located between q_1 and q_2 , we conclude q_1 and q_2 are like-sign. Consequently, $q_1/q_2 = 9.0$.

(b) Now we have

$$\frac{1}{4\pi\epsilon_0} \frac{|q_1|Q}{(-a-3a/2)^2} = \frac{1}{4\pi\epsilon_0} \frac{|q_2|Q}{(a-3a/2)^2}$$

which yields $|q_1| = 25 |q_2|$. Now, Q is not located between q_1 and q_2 ; one of them must push and the other must pull. Thus, they are unlike-sign, so $q_1/q_2 = -25$.

15. (a) The distance between q_1 and q_2 is

$$r_{12} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} = \sqrt{(-0.020 \text{ m} - 0.035 \text{ m})^2 + (0.015 \text{ m} - 0.005 \text{ m})^2} = 0.056 \text{ m}.$$

The magnitude of the force exerted by q_1 on q_2 is

$$F_{21} = k \frac{|q_1 q_2|}{r_{12}^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) (3.0 \times 10^{-6} \text{ C}) (4.0 \times 10^{-6} \text{ C})}{(0.056 \text{ m})^2} = 35 \text{ N}.$$

(b) The vector $\vec{F}_{21}$ is directed toward q_1 and makes an angle θ with the $+x$ axis, where

$$\theta = \tan^{-1} \left(\frac{y_2 - y_1}{x_2 - x_1} \right) = \tan^{-1} \left(\frac{1.5 \text{ cm} - 0.5 \text{ cm}}{-2.0 \text{ cm} - 3.5 \text{ cm}} \right) = -10.3^\circ \approx -10^\circ.$$

(c) Let the third charge be located at (x_3, y_3) , a distance r from q_2 . We note that q_1 , q_2 , and q_3 must be collinear; otherwise, an equilibrium position for any one of them would be impossible to find. Furthermore, we cannot place q_3 on the same side of q_2 where we also find q_1 , since in that region both forces (exerted on q_2 by q_3 and q_1) would be in the same direction (since q_2 is attracted to both of them). Thus, in terms of the angle found in part (a), we have $x_3 = x_2 - r \cos \theta$ and $y_3 = y_2 - r \sin \theta$ (which means $y_3 > y_2$ since θ is negative). The magnitude of force exerted on q_2 by q_3 is $F_{23} = k |q_2 q_3| / r^2$, which must equal that of the force exerted on it by q_1 (found in part (a)). Therefore,

$$k \frac{|q_2 q_3|}{r^2} = k \frac{|q_1 q_2|}{r_{12}^2} \Rightarrow r = r_{12} \sqrt{\frac{q_3}{q_1}} = 0.0645 \text{ m} = 6.45 \text{ cm}.$$

Consequently, $x_3 = x_2 - r \cos \theta = -2.0 \text{ cm} - (6.45 \text{ cm}) \cos(-10^\circ) = -8.4 \text{ cm}$,

(d) and $y_3 = y_2 - r \sin \theta = 1.5 \text{ cm} - (6.45 \text{ cm}) \sin(-10^\circ) = 2.7 \text{ cm}$.

16. (a) According to the graph, when q_3 is very close to q_1 (at which point we can consider the force exerted by particle 1 on 3 to dominate) there is a (large) force in the positive x direction. This is a repulsive force, then, so we conclude q_1 has the same sign as q_3 . Thus, q_3 is a positive-valued charge.

(b) Since the graph crosses zero and particle 3 is *between* the others, q_1 must have the same sign as q_2 , which means it is also positive-valued. We note that it crosses zero at $r = 0.020 \text{ m}$ (which is a distance $d = 0.060 \text{ m}$ from q_2). Using Coulomb's law at that point, we have

$$\frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q_3 q_2}{d^2} \Rightarrow q_2 = \left(\frac{d}{r} \right)^2 q_1 = \left(\frac{0.060 \text{ m}}{0.020 \text{ m}} \right)^2 q_1 = 9.0 q_1,$$

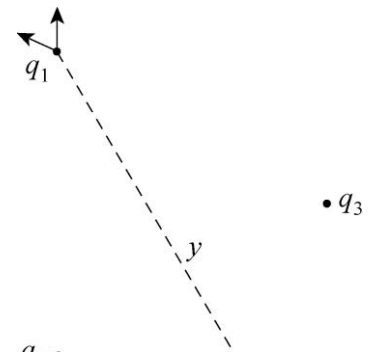
or $q_2/q_1 = 9.0$.

17. (a) Equation 21-1 gives

$$F_{12} = k \frac{q_1 q_2}{d^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) \frac{(20.0 \times 10^{-6} \text{ C})^2}{(1.50 \text{ m})^2} = 1.60 \text{ N}.$$

(b) On the right, a force diagram is shown as well as our choice of y axis (the dashed line).

The y axis is meant to bisect the line between q_2 and q_3 in order to make use of the symmetry in the problem (equilateral triangle of side length d , equal-magnitude charges $q_1 = q_2 = q_3 = q$). We see



that the resultant force is along this symmetry axis, and we obtain

$$|F_y| = 2 \left(k \frac{q^2}{d^2} \right) \cos 30^\circ = 2 \left(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2 \right) \frac{(20.0 \times 10^{-6} \text{ C})^2}{(1.50 \text{ m})^2} \cos 30^\circ = 2.77 \text{ N}.$$

18. Since the forces involved are proportional to q , we see that the essential difference between the two situations is $F_a \propto q_B + q_C$ (when those two charges are on the same side) versus $F_b \propto -q_B + q_C$ (when they are on opposite sides). Setting up ratios, we have

$$\frac{F_a}{F_b} = \frac{q_B + q_C}{-q_B + q_C} \Rightarrow \frac{2.014 \times 10^{-23} \text{ N}}{-2.877 \times 10^{-24} \text{ N}} = \frac{1 + q_C / q_B}{-1 + q_C / q_B}.$$

After noting that the ratio on the left hand side is very close to -7 , then, after a couple of algebra steps, we are led to

$$\frac{q_C}{q_B} = \frac{7+1}{7-1} = \frac{8}{6} = 1.333.$$

19. **THINK** Our system consists of two charges in a straight line. We'd like to place a third charge so that all three charges are in equilibrium.

EXPRESS If the system of three charges is to be in equilibrium, the force on each charge must be zero. The third charge q_3 must lie between the other two or else the forces acting on it due to the other charges would be in the same direction and q_3 could not be in equilibrium. Suppose q_3 is at a distance x from q , and $L - x$ from $4.00q$. The force acting on it is then given by

$$F_3 = \frac{1}{4\pi\epsilon_0} \left(\frac{qq_3}{x^2} - \frac{4qq_3}{(L-x)^2} \right)$$

where the positive direction is rightward. We require $F_3 = 0$ and solve for x .

ANALYZE (a) Canceling common factors yields $1/x^2 = 4/(L-x)^2$ and taking the square root yields $1/x = 2/(L-x)$. The solution is $x = L/3$. With $L = 9.00 \text{ cm}$, we have $x = 3.00 \text{ cm}$.

(b) Similarly, the y coordinate of q_3 is $y = 0$.

(c) The force on q is

$$F_q = \frac{-1}{4\pi\epsilon_0} \left(\frac{qq_3}{x^2} + \frac{4.00q^2}{L^2} \right).$$

The signs are chosen so that a negative force value would cause q to move leftward. We require $F_q = 0$ and solve for q_3 :

$$q_3 = -\frac{4qx^2}{L^2} = -\frac{4}{9}q \Rightarrow \frac{q_3}{q} = -\frac{4}{9} = -0.444$$

where $x = L/3$ is used.

LEARN We may also verify that the force on $4.00q$ also vanishes:

$$F_{4q} = \frac{1}{4\pi\epsilon_0} \left(\frac{4q^2}{L^2} + \frac{4qq_0}{(L-x)^2} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{4q^2}{L^2} + \frac{4(-4/9)q^2}{(4/9)L^2} \right) = \frac{1}{4\pi\epsilon_0} \left(\frac{4q^2}{L^2} - \frac{4q^2}{L^2} \right) = 0.$$

20. We note that the problem is examining the force on charge A , so that the respective distances (involved in the Coulomb force expressions) between B and A , and between C and A , do not change as particle B is moved along its circular path. We focus on the endpoints ($\theta = 0^\circ$ and 180°) of each graph, since they represent cases where the forces (on A) due to B and C are either parallel or anti-parallel (yielding maximum or minimum force magnitudes, respectively). We note, too, that since Coulomb's law is inversely proportional to r^2 then (if, say, the charges were all the same) the force due to C would be one-fourth as big as that due to B (since C is twice as far away from A). The charges, it turns out, are not the same, so there is also a factor of the charge ratio ξ (the charge of C divided by the charge of B), as well as the aforementioned $1/4$ factor. That is, the force exerted by C is, by Coulomb's law, equal to $\pm 1/4\xi$ multiplied by the force exerted by B .

(a) The maximum force is $2F_0$ and occurs when $\theta = 180^\circ$ (B is to the left of A , while C is the right of A). We choose the minus sign and write

$$2F_0 = (1 - 1/4\xi)F_0 \Rightarrow \xi = -4.$$

One way to think of the minus sign choice is $\cos(180^\circ) = -1$. This is certainly consistent with the minimum force ratio (zero) at $\theta = 0^\circ$ since that would also imply

$$0 = 1 + 1/4\xi \Rightarrow \xi = -4.$$

(b) The ratio of maximum to minimum forces is $1.25/0.75 = 5/3$ in this case, which implies

$$\frac{5}{3} = \frac{1 + 1/4\xi}{1 - 1/4\xi} \Rightarrow \xi = 16.$$

Of course, this could also be figured as illustrated in part (a), looking at the maximum force ratio by itself and solving, or looking at the minimum force ratio ($3/4$) at $\theta = 180^\circ$ and solving for ξ .

21. The charge dq within a thin shell of thickness dr is $dq = \rho dV = \rho A dr$ where $A = 4\pi r^2$. Thus, with $\rho = b/r$, we have

$$q = \int dq = 4\pi b \int_{r_1}^{r_2} r dr = 2\pi b [r_2^2 - r_1^2].$$

With $b = 3.0 \mu\text{C}/\text{m}^2$, $r_2 = 0.06 \text{ m}$, and $r_1 = 0.04 \text{ m}$, we obtain $q = 0.038 \mu\text{C} = 3.8 \times 10^{-8} \text{ C}$.

22. (a) We note that $\cos(30^\circ) = \frac{1}{2}\sqrt{3}$, so that the dashed line distance in the figure is $r = 2d/\sqrt{3}$. The net force on q_1 due to the two charges q_3 and q_4 (with $|q_3| = |q_4| = 1.60 \times 10^{-19} \text{ C}$) on the y axis has magnitude

$$2 \frac{|q_1 q_3|}{4\pi\epsilon_0 r^2} \cos(30^\circ) = \frac{3\sqrt{3}|q_1 q_3|}{16\pi\epsilon_0 d^2}.$$

This must be set equal to the magnitude of the force exerted on q_1 by $q_2 = 8.00 \times 10^{-19} \text{ C} = 5.00 |q_3|$ in order that its net force be zero:

$$\frac{3\sqrt{3}|q_1 q_3|}{16\pi\epsilon_0 d^2} = \frac{|q_1 q_2|}{4\pi\epsilon_0 (D+d)^2} \Rightarrow D = d \left(2\sqrt{\frac{5}{3\sqrt{3}}} - 1 \right) = 0.9245 d.$$

Given $d = 2.00 \text{ cm}$, this then leads to $D = 1.92 \text{ cm}$.

(b) As the angle decreases, its cosine increases, resulting in a larger contribution from the charges on the y axis. To offset this, the force exerted by q_2 must be made stronger, so that it must be brought closer to q_1 (keep in mind that Coulomb's law is *inversely* proportional to distance-squared). Thus, D must be decreased.

23. If θ is the angle between the force and the x -axis, then

$$\cos\theta = \frac{x}{\sqrt{x^2 + d^2}}.$$

We note that, due to the symmetry in the problem, there is no y component to the net force on the third particle. Thus, F represents the magnitude of force exerted by q_1 or q_2 on q_3 . Let $e = +1.60 \times 10^{-19} \text{ C}$, then $q_1 = q_2 = +2e$ and $q_3 = 4.0e$ and we have

$$F_{\text{net}} = 2F \cos\theta = \frac{2(2e)(4e)}{4\pi\epsilon_0 (x^2 + d^2)} \frac{x}{\sqrt{x^2 + d^2}} = \frac{4e^2 x}{\pi\epsilon_0 (x^2 + d^2)^{3/2}}.$$

(a) To find where the force is at an extremum, we can set the derivative of this expression equal to zero and solve for x , but it is good in any case to graph the function for a fuller understanding of its behavior, and as a quick way to see whether an extremum point is a maximum or a minimum. In this way, we find that the value coming from the derivative procedure is a maximum (and will be presented in part (b)) and that the minimum is found at the lower limit of the interval. Thus, the net force is found to be zero at $x = 0$, which is the smallest value of the net force in the interval $5.0 \text{ m} \geq x \geq 0$.

(b) The maximum is found to be at $x = d/\sqrt{2}$ or roughly 12 cm .

(c) The value of the net force at $x = 0$ is $F_{\text{net}} = 0$.

(d) The value of the net force at $x = d/\sqrt{2}$ is $F_{\text{net}} = 4.9 \times 10^{-26} \text{ N}$.

24. (a) Equation 21-1 gives

$$F = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.00 \times 10^{-16} \text{ C})^2}{(1.00 \times 10^{-2} \text{ m})^2} = 8.99 \times 10^{-19} \text{ N}.$$

(b) If n is the number of excess electrons (of charge $-e$ each) on each drop then

$$n = -\frac{q}{e} = -\frac{-1.00 \times 10^{-16} \text{ C}}{1.60 \times 10^{-19} \text{ C}} = 625.$$

25. Equation 21-11 (in absolute value) gives $n = \frac{|q|}{e} = \frac{1.0 \times 10^{-7} \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 6.3 \times 10^{11}$.

26. The magnitude of the force is

$$F = k \frac{e^2}{r^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(1.60 \times 10^{-19} \text{ C})^2}{(2.82 \times 10^{-10} \text{ m})^2} = 2.89 \times 10^{-9} \text{ N}.$$

27. **THINK** The magnitude of the electrostatic force between two charges q_1 and q_2 separated by a distance r is given by Coulomb's law.

EXPRESS Let the charge of the ions be q . With $q_1 = q_2 = +q$, the magnitude of the force between the (positive) ions is given by

$$F = \frac{qq}{4\pi\epsilon_0 r^2} = k \frac{q^2}{r^2},$$

where $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$.

ANALYZE (a) We solve for the charge:

$$q = r \sqrt{\frac{F}{k}} = (5.0 \times 10^{-10} \text{ m}) \sqrt{\frac{3.7 \times 10^{-9} \text{ N}}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2}} = 3.2 \times 10^{-19} \text{ C}.$$

(b) Let n be the number of electrons missing from each ion. Then, $ne = q$, or

$$n = \frac{q}{e} = \frac{3.2 \times 10^{-9} \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 2.$$

LEARN Electric charge is quantized. This means that any charge can be written as $q = ne$, where n is an integer (positive or negative), and $e = 1.6 \times 10^{-19} \text{ C}$ is the elementary charge.

28. Keeping in mind that an ampere is a coulomb per second ($1 \text{ A} = 1 \text{ C/s}$), and that a minute is 60 seconds, the charge (in absolute value) that passes through the chest is

$$|q| = (0.300 \text{ C/s})(120 \text{ s}) = 36.0 \text{ C}.$$

This charge consists of n electrons (each of which has an absolute value of charge equal to e). Thus,

$$n = \frac{|q|}{e} = \frac{36.0 \text{ C}}{1.60 \times 10^{-19} \text{ C}} = 2.25 \times 10^{20}.$$

29. (a) We note that $\tan(30^\circ) = 1/\sqrt{3}$. In the initial (highly symmetrical) configuration, the net force on the central bead is in the $-y$ direction and has magnitude $3F$ where F is the Coulomb's law force of one bead on another at distance $d = 10 \text{ cm}$. This is due to the fact that the forces exerted on the central bead (in the initial situation) by the beads on the x axis cancel each other; also, the force exerted "downward" by bead 4 on the central bead is four times larger than the "upward" force exerted by bead 2. This net force along the y axis does not change as bead 1 is now moved, though there is now a nonzero x -component F_x . The components are now related by

$$\tan(30^\circ) = \frac{F_x}{F_y} \Rightarrow \frac{1}{\sqrt{3}} = \frac{F_x}{3F}$$

which implies $F_x = \sqrt{3} F$. Now, bead 3 exerts a "leftward" force of magnitude F on the central bead, while bead 1 exerts a "rightward" force of magnitude F' . Therefore,

$$F' - F = \sqrt{3} F. \quad \Rightarrow \quad F' = (\sqrt{3} + 1) F.$$

The fact that Coulomb's law depends inversely on distance-squared then implies

$$r^2 = \frac{d^2}{\sqrt{3} + 1} \Rightarrow r = \frac{d}{\sqrt{\sqrt{3} + 1}} = \frac{10 \text{ cm}}{\sqrt{\sqrt{3} + 1}} = \frac{10 \text{ cm}}{1.65} = 6.05 \text{ cm}$$

where r is the distance between bead 1 and the central bead. This corresponds to $x = -6.05 \text{ cm}$.

(b) To regain the condition of high symmetry (in particular, the cancellation of x -components) bead 3 must be moved closer to the central bead so that it, too, is the distance r (as calculated in part (a)) away from it.

30. (a) Let x be the distance between particle 1 and particle 3. Thus, the distance between particle 3 and particle 2 is $L - x$. Both particles exert leftward forces on q_3 (so long as it is on the line between them), so the magnitude of the net force on q_3 is

$$F_{\text{net}} = |F_{13}| + |F_{23}| = \frac{|q_1 q_3|}{4\pi\epsilon_0 x^2} + \frac{|q_2 q_3|}{4\pi\epsilon_0 (L-x)^2} = \frac{e^2}{\pi\epsilon_0} \left(\frac{1}{x^2} + \frac{27}{(L-x)^2} \right)$$

with the values of the charges (stated in the problem) plugged in. Finding the value of x that minimizes this expression leads to $x = \frac{1}{4}L$. Thus, $x = 2.00$ cm.

(b) Substituting $x = \frac{1}{4}L$ back into the expression for the net force magnitude and using the standard value for e leads to $F_{\text{net}} = 9.21 \times 10^{-24}$ N.

31. The unit ampere is discussed in Section 21-4. The proton flux is given as 1500 protons per square meter per second, where each proton provides a charge of $q = +e$. The current through the spherical area $4\pi R^2 = 4\pi (6.37 \times 10^6 \text{ m})^2 = 5.1 \times 10^{14} \text{ m}^2$ would be

$$i = (5.1 \times 10^{14} \text{ m}^2) \left(1500 \frac{\text{protons}}{\text{s} \cdot \text{m}^2} \right) (1.6 \times 10^{-19} \text{ C/proton}) = 0.122 \text{ A}.$$

32. Since the graph crosses zero, q_1 must be positive-valued: $q_1 = +8.00e$. We note that it crosses zero at $r = 0.40$ m. Now the asymptotic value of the force yields the magnitude and sign of q_2 :

$$\frac{q_1 q_2}{4\pi\epsilon_0 r^2} = F \Rightarrow q_2 = \left(\frac{1.5 \times 10^{-25}}{k q_1} \right) r^2 = 2.086 \times 10^{-18} \text{ C} = 13e.$$

33. The volume of 250 cm^3 corresponds to a mass of 250 g since the density of water is 1.0 g/cm^3 . This mass corresponds to $250/18 = 14$ moles since the molar mass of water is 18. There are ten protons (each with charge $q = +e$) in each molecule of H_2O , so

$$Q = 14 N_A q = 14 (6.02 \times 10^{23}) (10) (1.60 \times 10^{-19} \text{ C}) = 1.3 \times 10^7 \text{ C}.$$

34. Let d be the vertical distance from the coordinate origin to $q_3 = -q$ and $q_4 = -q$ on the $+y$ axis, where the symbol q is assumed to be a positive value. Similarly, d is the (positive) distance from the origin $q_4 = -$ on the $-y$ axis. If we take each angle θ in the figure to be positive, then we have

$$\tan \theta = d/R \text{ and } \cos \theta = R/r,$$

where r is the dashed line distance shown in the figure. The problem asks us to consider θ to be a variable in the sense that, once the charges on the x axis are fixed in place (which determines R), d can then be arranged to some multiple of R , since $d = R \tan \theta$. The aim of this exploration is to show that if q is bounded then θ (and thus d) is also bounded.

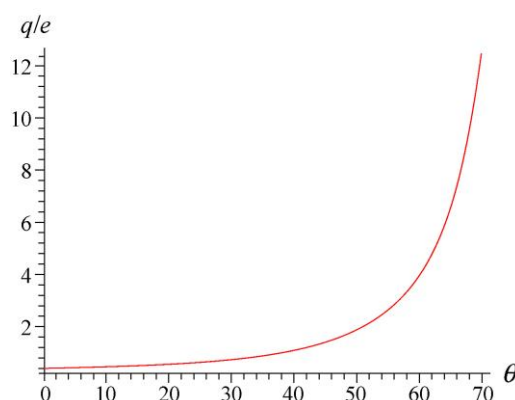
From symmetry, we see that there is no net force in the vertical direction on $q_2 = -e$ sitting at a distance R to the left of the coordinate origin. We note that the net x force caused by q_3 and q_4 on the y -axis will have a magnitude equal to

$$2 \frac{qe}{4\pi\epsilon_0 r^2} \cos \theta = \frac{2qe \cos \theta}{4\pi\epsilon_0 (R/\cos \theta)^2} = \frac{2qe \cos^3 \theta}{4\pi\epsilon_0 R^2}.$$

Consequently, to achieve a zero net force along the x axis, the above expression must equal the magnitude of the repulsive force exerted on q_2 by $q_1 = -e$. Thus,

$$\frac{2qe \cos^3 \theta}{4\pi\epsilon_0 R^2} = \frac{e^2}{4\pi\epsilon_0 R^2} \Rightarrow q = \frac{e}{2 \cos^3 \theta}.$$

Below we plot q/e as a function of the angle (in degrees):



The graph suggests that $q/e < 5$ for $\theta < 60^\circ$, roughly. We can be more precise by solving the above equation. The requirement that $q \leq 5e$ leads to

$$\frac{e}{2 \cos^3 \theta} \leq 5e \Rightarrow \frac{1}{(10)^{1/3}} \leq \cos \theta$$

which yields $\theta \leq 62.34^\circ$. The problem asks for “physically possible values,” and it is reasonable to suppose that only positive-integer-multiple values of e are allowed for q . If we let $q = ne$, for $n = 1 \dots 5$, then θ_N will be found by taking the inverse cosine of the cube root of $(1/2n)$.

- (a) The smallest value of angle is $\theta_1 = 37.5^\circ$ (or 0.654 rad).
- (b) The second smallest value of angle is $\theta_2 = 50.95^\circ$ (or 0.889 rad).
- (c) The third smallest value of angle is $\theta_3 = 56.6^\circ$ (or 0.988 rad).

35. THINK Our system consists of 8 Cs^+ ions at the corners of a cube and a Cl^- ion at the cube's center. To calculate the electrostatic force on the Cl^- ion, we apply the superposition principle and make use of the symmetry property of the configuration.

EXPRESS In (a) where all 8 Cs^+ ions are present, every cesium ion at a corner of the cube exerts a force of the same magnitude on the chlorine ion at the cube center. Each force is attractive and is directed toward the cesium ion that exerts it, along the body diagonal of the cube. We can pair every cesium ion with another, diametrically positioned at the opposite corner of the cube.

In (b) where one Cs^+ ion is missing at the corner, rather than remove a cesium ion, we superpose charge $-e$ at the position of one cesium ion. This neutralizes the ion, and as far as the electrical force on the chlorine ion is concerned, it is equivalent to removing the ion. The forces of the eight cesium ions at the cube corners sum to zero, so the only force on the chlorine ion is the force of the added charge.

ANALYZE (a) Since the two Cs^+ ions in such a pair exert forces that have the same magnitude but are oppositely directed, the two forces sum to zero and, since every cesium ion can be paired in this way, the total force on the chlorine ion is zero.

(b) The length of a body diagonal of a cube is $\sqrt{3}a$, where a is the length of a cube edge. Thus, the distance from the center of the cube to a corner is $d = \frac{\sqrt{3}}{2}a$. The force has magnitude

$$F = k \frac{e^2}{d^2} = \frac{ke^2}{\frac{3}{4}a^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.60 \times 10^{-19} \text{ C})^2}{\frac{3}{4}(0.40 \times 10^{-9} \text{ m})^2} = 1.9 \times 10^{-9} \text{ N}.$$

Since both the added charge and the chlorine ion are negative, the force is one of repulsion. The chlorine ion is pushed away from the site of the missing cesium ion.

LEARN When solving electrostatic problems involving a discrete number of charges, symmetry argument can often be applied to simplify the problem.

36. (a) Since the proton is positively charged, the emitted particle must be a positron (as opposed to the negatively charged electron) in accordance with the law of charge conservation.

(b) In this case, the initial state had zero charge (the neutron is neutral), so the sum of charges in the final state must be zero. Since there is a proton in the final state, there should also be an electron (as opposed to a positron) so that $\Sigma q = 0$.

37. **THINK** Charges are conserved in nuclear reactions.

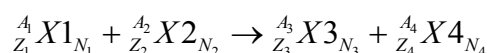
EXPRESS We note that none of the reactions given include a beta decay (see Chapter 42), so the number of protons (Z), the number of neutrons (N), and the number of electrons are each conserved. The mass number (total number of nucleons) is defined as $A = N + Z$. Atomic numbers (number of protons) and molar masses can be found in Appendix F of the text.

ANALYZE (a) ^1H has 1 proton, 1 electron, and 0 neutrons and ^9Be has 4 protons, 4 electrons, and $9 - 4 = 5$ neutrons, so X has $1 + 4 = 5$ protons, $1 + 4 = 5$ electrons, and $0 + 5 - 1 = 4$ neutrons. One of the neutrons is freed in the reaction. X must be boron with a molar mass of $5 + 4 = 9$ g/mol: ^9B .

(b) ^{12}C has 6 protons, 6 electrons, and $12 - 6 = 6$ neutrons and ^1H has 1 proton, 1 electron, and 0 neutrons, so X has $6 + 1 = 7$ protons, $6 + 1 = 7$ electrons, and $6 + 0 = 6$ neutrons. It must be nitrogen with a molar mass of $7 + 6 = 13$ g/mol: ^{13}N .

(c) ^{15}N has 7 protons, 7 electrons, and $15 - 7 = 8$ neutrons; ^1H has 1 proton, 1 electron, and 0 neutrons; and ^4He has 2 protons, 2 electrons, and $4 - 2 = 2$ neutrons; so X has $7 + 1 - 2 = 6$ protons, 6 electrons, and $8 + 0 - 2 = 6$ neutrons. It must be carbon with a molar mass of $6 + 6 = 12$ g/mol: ^{12}C .

LEARN A general expression for the reaction can be expressed as



where $A_i = Z_i + N_i$. Since the number of protons (Z), the number of neutrons (N), and the number of nucleons (A) are each conserved, we have $A_1 + A_2 = A_3 + A_4$, $Z_1 + Z_2 = Z_3 + Z_4$ and $N_1 + N_2 = N_3 + N_4$.

38. As a result of the first action, both sphere W and sphere A possess charge $\frac{1}{2}q_A$, where q_A is the initial charge of sphere A . As a result of the second action, sphere W has charge

$$\frac{1}{2} \left(\frac{q_A}{2} - 32e \right).$$

As a result of the final action, sphere W now has charge equal to

$$\frac{1}{2} \left[\frac{1}{2} \left(\frac{q_A}{2} - 32e \right) + 48e \right].$$

Setting this final expression equal to $+18e$ as required by the problem leads (after a couple of algebra steps) to the answer: $q_A = +16e$.

39. THINK We have two discrete charges in the xy -plane. The electrostatic force on particle 2 due to particle 1 has both x and y components.

EXPRESS Using Coulomb's law, the magnitude of the force of particle 1 on particle 2 is $F_{21} = k \frac{q_1 q_2}{r^2}$, where $r = \sqrt{d_1^2 + d_2^2}$ and $k = 1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2$. Since both q_1 and q_2 are positively charged, particle 2 is repelled by particle 1, so the direction of $\vec{F}_{21}$ is away from particle 1 and toward 2. In unit-vector notation, $\vec{F}_{21} = F_{21}\hat{r}$, where

$$\hat{r} = \frac{\vec{r}}{r} = \frac{d_2\hat{i} - d_1\hat{j}}{\sqrt{d_1^2 + d_2^2}}.$$

The x component of $\vec{F}_{21}$ is $F_{21,x} = F_{21}d_2 / \sqrt{d_1^2 + d_2^2}$.

ANALYZE Combining the expressions above, we obtain

$$\begin{aligned} F_{21,x} &= k \frac{q_1 q_2 d_2}{r^3} = k \frac{q_1 q_2 d_2}{(d_1^2 + d_2^2)^{3/2}} \\ &= \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(4 \cdot 1.60 \times 10^{-19} \text{ C})(6 \cdot 1.60 \times 10^{-19} \text{ C})(6.00 \times 10^{-3} \text{ m})}{[(2.00 \times 10^{-3} \text{ m})^2 + (6.00 \times 10^{-3} \text{ m})^2]^{3/2}} \\ &= 1.31 \times 10^{-22} \text{ N} \end{aligned}$$

LEARN In a similar manner, we find the y component of $\vec{F}_{21}$ to be

$$\begin{aligned} F_{21,y} &= -k \frac{q_1 q_2 d_1}{r^3} = -k \frac{q_1 q_2 d_1}{(d_1^2 + d_2^2)^{3/2}} \\ &= -\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(4 \cdot 1.60 \times 10^{-19} \text{ C})(6 \cdot 1.60 \times 10^{-19} \text{ C})(2.00 \times 10^{-3} \text{ m})}{[(2.00 \times 10^{-3} \text{ m})^2 + (6.00 \times 10^{-3} \text{ m})^2]^{3/2}} \\ &= -0.437 \times 10^{-22} \text{ N}. \end{aligned}$$

Thus, $\vec{F}_{21} = (1.31 \times 10^{-22} \text{ N})\hat{i} - (0.437 \times 10^{-22} \text{ N})\hat{j}$.

40. Regarding the forces on q_3 exerted by q_1 and q_2 , one must “push” and the other must “pull” in order that the net force is zero; hence, q_1 and q_2 have opposite signs. For individual forces to cancel, their magnitudes must be equal:

$$k \frac{|q_1| |q_3|}{(L_{12} + L_{23})^2} = k \frac{|q_2| |q_3|}{(L_{23})^2}.$$

With $L_{23} = 2.00L_{12}$, the above expression simplifies to $\frac{|q_1|}{9} = \frac{|q_2|}{4}$. Therefore, $q_1 = -9q_2/4$, or $q_1/q_2 = -2.25$.

41. (a) The magnitudes of the gravitational and electrical forces must be the same:

$$\frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2} = G \frac{mM}{r^2}$$

where q is the charge on either body, r is the center-to-center separation of Earth and Moon, G is the universal gravitational constant, M is the mass of Earth, and m is the mass of the Moon. We solve for q :

$$q = \sqrt{4\pi\epsilon_0 GmM}.$$

According to Appendix C of the text, $M = 5.98 \times 10^{24}$ kg, and $m = 7.36 \times 10^{22}$ kg, so (using $4\pi\epsilon_0 = 1/k$) the charge is

$$q = \sqrt{\frac{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(7.36 \times 10^{22} \text{ kg})(5.98 \times 10^{24} \text{ kg})}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2}} = 5.7 \times 10^{13} \text{ C}.$$

(b) The distance r cancels because both the electric and gravitational forces are proportional to $1/r^2$.

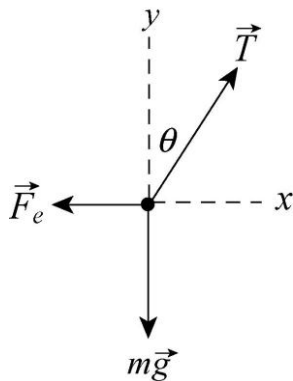
(c) The charge on a hydrogen ion is $e = 1.60 \times 10^{-19}$ C, so there must be

$$n = \frac{q}{e} = \frac{5.7 \times 10^{13} \text{ C}}{1.6 \times 10^{-19} \text{ C}} = 3.6 \times 10^{32} \text{ ions}.$$

Each ion has a mass of $m_i = 1.67 \times 10^{-27}$ kg, so the total mass needed is

$$m = nm_i = (3.6 \times 10^{32})(1.67 \times 10^{-27} \text{ kg}) = 6.0 \times 10^5 \text{ kg}.$$

42. (a) A force diagram for one of the balls is shown below. The force of gravity $m\vec{g}$ acts downward, the electrical force $\vec{F}_e$ of the other ball acts to the left, and the tension in the thread acts along the thread, at the angle θ to the vertical. The ball is in equilibrium, so its acceleration is zero. The y component of Newton's second law yields $T \cos \theta - mg = 0$ and the x component yields $T \sin \theta - F_e = 0$. We solve the first equation for T and obtain $T = mg/\cos \theta$. We substitute the result into the second to obtain $mg \tan \theta - F_e = 0$.



Examination of the geometry of the figure shown leads to $\tan \theta = \frac{x/2}{\sqrt{L^2 - (x/2)^2}}$.

If L is much larger than x (which is the case if θ is very small), we may neglect $x/2$ in the denominator and write $\tan \theta \approx x/2L$. This is equivalent to approximating $\tan \theta$ by $\sin \theta$. The magnitude of the electrical force of one ball on the other is

$$F_e = \frac{q^2}{4\pi\epsilon_0 x^2}$$

by Eq. 21-4. When these two expressions are used in the equation $mg \tan \theta = F_e$, we obtain

$$\frac{mgx}{2L} \approx \frac{1}{4\pi\epsilon_0} \frac{q^2}{x^2} \Rightarrow x \approx \left(\frac{q^2 L}{2\pi\epsilon_0 mg} \right)^{1/3}.$$

(b) We solve $x^3 = 2kq^2L/mg$ for the charge (using Eq. 21-5):

$$q = \sqrt{\frac{mgx^3}{2kL}} = \sqrt{\frac{(0.010\text{ kg})(9.8\text{ m/s}^2)(0.050\text{ m})^3}{2(8.99 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)(1.20\text{ m})}} = \pm 2.4 \times 10^{-8} \text{ C}.$$

Thus, the magnitude is $|q| = 2.4 \times 10^{-8} \text{ C}$.

43. (a) If one of them is discharged, there would be no electrostatic repulsion between the two balls and they would both come to the position $\theta = 0$, making contact with each other.

(b) A redistribution of the remaining charge would then occur, with each of the balls getting $q/2$. Then they would again be separated due to electrostatic repulsion, which results in the new equilibrium separation

$$x' = \left[\frac{(q/2)^2 L}{2\pi\epsilon_0 mg} \right]^{1/3} = \left(\frac{1}{4} \right)^{1/3} x = \left(\frac{1}{4} \right)^{1/3} (5.0 \text{ cm}) = 3.1 \text{ cm}.$$

44. **THINK** The problem compares the electrostatic force between two protons and the gravitational force by Earth on a proton.

EXPRESS The magnitude of the gravitational force on a proton near the surface of the Earth is $F_g = mg$, where $m = 1.67 \times 10^{-27} \text{ kg}$ is the mass of the proton. On the other hand, the electrostatic force between two protons separated by a distance r is $F_e = kq^2 / r$. When the two forces are equal, we have $kq^2 / r^2 = mg$.

ANALYZE Solving for r , we obtain

$$r = q \sqrt{\frac{k}{mg}} = (1.60 \times 10^{-19} \text{ C}) \sqrt{\frac{8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2}{(1.67 \times 10^{-27} \text{ kg})(9.8 \text{ m/s}^2)}} = 0.119 \text{ m}.$$

LEARN The electrostatic force at this distance is $F_e = F_g = 1.64 \times 10^{-26} \text{ N}$.

45. There are two protons (each with charge $q = +e$) in each molecule, so

$$Q = N_A q = (6.02 \times 10^{23}) (2)(1.60 \times 10^{-19} \text{ C}) = 1.9 \times 10^5 \text{ C} = 0.19 \text{ MC}.$$

46. Let $\vec{F}_{12}$ denotes the force on q_1 exerted by q_2 and F_{12} be its magnitude.

(a) We consider the net force on q_1 . $\vec{F}_{12}$ points in the $+x$ direction since q_1 is attracted to q_2 . $\vec{F}_{13}$ and $\vec{F}_{14}$ both point in the $-x$ direction since q_1 is repelled by q_3 and q_4 . Thus, using $d = 0.0200 \text{ m}$, the net force is

$$\begin{aligned} F_1 = F_{12} - F_{13} - F_{14} &= \frac{2e|-e|}{4\pi\epsilon_0 d^2} - \frac{(2e)(e)}{4\pi\epsilon_0 (2d)^2} - \frac{(2e)(4e)}{4\pi\epsilon_0 (3d)^2} = \frac{11}{18} \frac{e^2}{4\pi\epsilon_0 d^2} \\ &= \frac{11}{18} \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(1.60 \times 10^{-19} \text{ C})^2}{(2.00 \times 10^{-2} \text{ m})^2} = 3.52 \times 10^{-25} \text{ N} \end{aligned}$$

or $\vec{F}_1 = (3.52 \times 10^{-25} \text{ N})\hat{i}$.

(b) We now consider the net force on q_2 . We note that $\vec{F}_{21} = -\vec{F}_{12}$ points in the $-x$ direction, and $\vec{F}_{23}$ and $\vec{F}_{24}$ both point in the $+x$ direction. The net force is

$$F_{23} + F_{24} - F_{21} = \frac{4e|-e|}{4\pi\epsilon_0(2d)^2} + \frac{e|-e|}{4\pi\epsilon_0 d^2} - \frac{2e|-e|}{4\pi\epsilon_0 d^2} = 0.$$

47. We are looking for a charge q that, when placed at the origin, experiences $\vec{F}_{\text{net}} = 0$, where

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3.$$

The magnitude of these individual forces are given by Coulomb's law, Eq. 21-1, and without loss of generality we assume $q > 0$. The charges q_1 ($+6 \mu\text{C}$), q_2 ($-4 \mu\text{C}$), and q_3 (unknown), are located on the $+x$ axis, so that we know $\vec{F}_1$ points toward $-x$, $\vec{F}_2$ points toward $+x$, and $\vec{F}_3$ points toward $-x$ if $q_3 > 0$ and points toward $+x$ if $q_3 < 0$. Therefore, with $r_1 = 8 \text{ m}$, $r_2 = 16 \text{ m}$ and $r_3 = 24 \text{ m}$, we have

$$0 = -k \frac{q_1 q}{r_1^2} + k \frac{|q_2| q}{r_2^2} - k \frac{q_3 q}{r_3^2}.$$

Simplifying, this becomes

$$0 = -\frac{6}{8^2} + \frac{4}{16^2} - \frac{q_3}{24^2}$$

where q_3 is now understood to be in μC . Thus, we obtain $q_3 = -45 \mu\text{C}$.

48. (a) Since $q_A = -2.00 \text{ nC}$ and $q_C = +8.00 \text{ nC}$, Eq. 21-4 leads to

$$|\vec{F}_{AC}| = \frac{|q_A q_C|}{4\pi\epsilon_0 d^2} = \frac{|(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(-2.00 \times 10^{-9} \text{ C})(8.00 \times 10^{-9} \text{ C})|}{(0.200 \text{ m})^2} = 3.60 \times 10^{-6} \text{ N}.$$

(b) After making contact with each other, both A and B have a charge of

$$\frac{q_A + q_B}{2} = \left(\frac{-2.00 + (-4.00)}{2} \right) \text{ nC} = -3.00 \text{ nC}.$$

When B is grounded its charge is zero. After making contact with C , which has a charge of $+8.00 \text{ nC}$, B acquires a charge of $[0 + (-8.00 \text{ nC})]/2 = -4.00 \text{ nC}$, which charge C has as well. Finally, we have $Q_A = -3.00 \text{ nC}$ and $Q_B = Q_C = -4.00 \text{ nC}$. Therefore,

$$|\vec{F}_{AC}| = \frac{|q_A q_C|}{4\pi\epsilon_0 d^2} = \frac{|(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(-3.00 \times 10^{-9} \text{ C})(-4.00 \times 10^{-9} \text{ C})|}{(0.200 \text{ m})^2} = 2.70 \times 10^{-6} \text{ N}.$$

(c) We also obtain

$$|\vec{F}_{BC}| = \frac{|q_B q_C|}{4\pi\epsilon_0 d^2} = \frac{|(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(-4.00 \times 10^{-9} \text{ C})(-4.00 \times 10^{-9} \text{ C})|}{(0.200 \text{ m})^2} = 3.60 \times 10^{-6} \text{ N}.$$

49. Coulomb's law gives

$$F = \frac{|q|^2}{4\pi\epsilon_0 r^2} = \frac{k(e/3)^2}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.60 \times 10^{-19} \text{ C})^2}{9(2.6 \times 10^{-15} \text{ m})^2} = 3.8 \text{ N}.$$

50. (a) Since the rod is in equilibrium, the net force acting on it is zero, and the net torque about any point is also zero. We write an expression for the net torque about the bearing, equate it to zero, and solve for x . The charge Q on the left exerts an upward force of magnitude $(1/4\pi\epsilon_0)(qQ/h^2)$, at a distance $L/2$ from the bearing. We take the torque to be negative. The attached weight exerts a downward force of magnitude W , at a distance $x - L/2$ from the bearing. This torque is also negative. The charge Q on the right exerts an upward force of magnitude $(1/4\pi\epsilon_0)(2qQ/h^2)$, at a distance $L/2$ from the bearing. This torque is positive. The equation for rotational equilibrium is

$$\frac{-1}{4\pi\epsilon_0} \frac{qQ}{h^2} \frac{L}{2} - W \left(x - \frac{L}{2} \right) + \frac{1}{4\pi\epsilon_0} \frac{2qQ}{h^2} \frac{L}{2} = 0.$$

The solution for x is

$$x = \frac{L}{2} \left(1 + \frac{1}{4\pi\epsilon_0} \frac{qQ}{h^2 W} \right).$$

(b) If F_N is the magnitude of the upward force exerted by the bearing, then Newton's second law (with zero acceleration) gives

$$W - \frac{1}{4\pi\epsilon_0} \frac{qQ}{h^2} - \frac{1}{4\pi\epsilon_0} \frac{2qQ}{h^2} - F_N = 0.$$

We solve for h so that $F_N = 0$. The result is

$$h = \sqrt{\frac{1}{4\pi\epsilon_0} \frac{3qQ}{W}}.$$

51. The charge dq within a thin section of the rod (of thickness dx) is $\rho A dx$ where $A = 4.00 \times 10^{-4} \text{ m}^2$ and ρ is the charge per unit volume. The number of (excess) electrons in the rod (of length $L = 2.00 \text{ m}$) is $n = q/(-e)$ where e is given in Eq. 21-12.

(a) In the case where $\rho = -4.00 \times 10^{-6} \text{ C/m}^3$, we have

$$n = \frac{q}{-e} = \frac{\rho A}{-e} \int_0^L dx = \frac{|\rho| AL}{e} = 2.00 \times 10^{10}.$$

(b) With $\rho = bx^2$ ($b = -2.00 \times 10^{-6} \text{ C/m}^5$) we obtain

$$n = \frac{b A}{-e} \int_0^L x^2 dx = \frac{|b| AL^3}{3e} = 1.33 \times 10^{10}.$$

52. For the Coulomb force to be sufficient for circular motion at that distance (where $r = 0.200 \text{ m}$ and the acceleration needed for circular motion is $a = v^2/r$) the following equality is required:

$$\frac{Qq}{4\pi\epsilon_0 r^2} = -\frac{mv^2}{r}.$$

With $q = 4.00 \times 10^{-6} \text{ C}$, $m = 0.000800 \text{ kg}$, $v = 50.0 \text{ m/s}$, this leads to

$$Q = -\frac{4\pi\epsilon_0 r m v^2}{q} = -\frac{(0.200 \text{ m})(8.00 \times 10^{-4} \text{ kg})(50.0 \text{ m/s})^2}{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(4.00 \times 10^{-6} \text{ C})} = -1.11 \times 10^{-5} \text{ C}.$$

53. (a) Using Coulomb's law, we obtain

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} = \frac{kq^2}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.00 \text{ C})^2}{(1.00 \text{ m})^2} = 8.99 \times 10^9 \text{ N}.$$

(b) If $r = 1000 \text{ m}$, then

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} = \frac{kq^2}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.00 \text{ C})^2}{(1.00 \times 10^3 \text{ m})^2} = 8.99 \times 10^3 \text{ N}.$$

54. Let q_1 be the charge of one part and q_2 that of the other part; thus, $q_1 + q_2 = Q = 6.0 \text{ } \mu\text{C}$. The repulsive force between them is given by Coulomb's law:

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 r^2} = \frac{q_1(Q - q_1)}{4\pi\epsilon_0 r^2}.$$

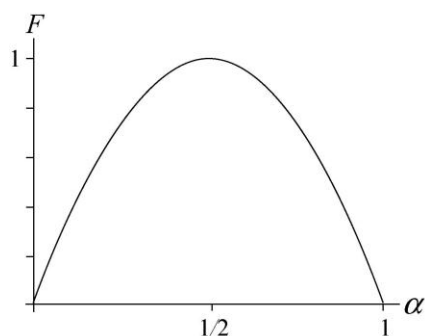
If we maximize this expression by taking the derivative with respect to q_1 and setting equal to zero, we find $q_1 = Q/2$, which might have been anticipated (based on symmetry arguments). This implies $q_2 = Q/2$ also. With $r = 0.0030 \text{ m}$ and $Q = 6.0 \times 10^{-6} \text{ C}$, we find

$$F = \frac{(Q/2)(Q/2)}{4\pi\epsilon_0 r^2} = \frac{1}{4} \frac{Q^2}{4\pi\epsilon_0 r^2} = \frac{1}{4} \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(6.0 \times 10^{-6} \text{ C})^2}{(3.00 \times 10^{-3} \text{ m})^2} \approx 9.0 \times 10^3 \text{ N}.$$

55. The two charges are $q = \alpha Q$ (where α is a pure number presumably less than 1 and greater than zero) and $Q - q = (1 - \alpha)Q$. Thus, Eq. 21-4 gives

$$F = \frac{1}{4\pi\epsilon_0} \frac{|\alpha Q q| |1 - \alpha| Q}{d^2} = \frac{Q^2 \alpha |1 - \alpha|}{4\pi\epsilon_0 d^2}.$$

The graph below, of F versus α , has been scaled so that the maximum is 1. In actuality, the maximum value of the force is $F_{\max} = Q^2/16\pi\epsilon_0 d^2$.



(a) It is clear that $\alpha = 1/2 = 0.5$ gives the maximum value of F .

(b) Seeking the half-height points on the graph is difficult without grid lines or some of the special tracing features found in a variety of modern calculators. It is not difficult to algebraically solve for the half-height points (this involves the use of the quadratic formula). The results are

$$\alpha_1 = \frac{1}{2} \left(1 - \frac{1}{\sqrt{2}} \right) \approx 0.15 \quad \text{and} \quad \alpha_2 = \frac{1}{2} \left(1 + \frac{1}{\sqrt{2}} \right) \approx 0.85.$$

Thus, the smaller value of α is $\alpha_1 = 0.15$,

(c) and the larger value of α is $\alpha_2 = 0.85$.

56. (a) Equation 21-11 (in absolute value) gives

$$n = \frac{|q|}{e} = \frac{2.00 \times 10^{-6} \text{ C}}{1.60 \times 10^{-19} \text{ C}} = 1.25 \times 10^{13} \text{ electrons}.$$

(b) Since you have the excess electrons (and electrons are lighter and more mobile than protons) then the electrons “leap” from you to the faucet instead of protons moving from the faucet to you (in the process of neutralizing your body).

(c) Unlike charges attract, and the faucet (which is grounded and is able to gain or lose any number of electrons due to its contact with Earth’s large reservoir of mobile charges) becomes positively charged, especially in the region closest to your (negatively charged) hand, just before the spark.

(d) The cat is positively charged (before the spark), and by the reasoning given in part (b) the flow of charge (electrons) is from the faucet to the cat.

(e) If we think of the nose as a conducting sphere, then the side of the sphere closest to the fur is of one sign (of charge) and the side furthest from the fur is of the opposite sign (which, additionally, is oppositely charged from your bare hand, which had stroked the cat’s fur). The charges in your hand and those of the furthest side of the “sphere” therefore attract each other, and when close enough, manage to neutralize (due to the “jump” made by the electrons) in a painful spark.

57. If the relative difference between the proton and electron charges (in absolute value) were

$$\frac{q_p - |q_e|}{e} = 0.0000010$$

then the actual difference would be $q_p - |q_e| = 1.6 \times 10^{-25} \text{ C}$. Amplified by a factor of $29 \times 3 \times 10^{22}$ as indicated in the problem, this amounts to a deviation from perfect neutrality of

$$\Delta q = (29 \times 3 \times 10^{22})(1.6 \times 10^{-25} \text{ C}) = 0.14 \text{ C}$$

in a copper penny. Two such pennies, at $r = 1.0 \text{ m}$, would therefore experience a very large force. Equation 21-1 gives

$$F = k \frac{(\Delta q)^2}{r^2} = 1.7 \times 10^8 \text{ N}.$$

58. Charge $q_1 = -80 \times 10^{-6} \text{ C}$ is at the origin, and charge $q_2 = +40 \times 10^{-6} \text{ C}$ is at $x = 0.20 \text{ m}$. The force on $q_3 = +20 \times 10^{-6} \text{ C}$ is due to the attractive and repulsive forces from q_1 and q_2 , respectively. In symbols, $\vec{F}_{3 \text{ net}} = \vec{F}_{31} + \vec{F}_{32}$, where

$$|\vec{F}_{31}| = k \frac{q_3 |q_1|}{r_{31}^2}, \quad |\vec{F}_{32}| = k \frac{q_3 q_2}{r_{32}^2}.$$

(a) In this case $r_{31} = 0.40 \text{ m}$ and $r_{32} = 0.20 \text{ m}$, with $\vec{F}_{31}$ directed toward $-x$ and $\vec{F}_{32}$ directed in the $+x$ direction. Using the value of k in Eq. 21-5, we obtain

$$\begin{aligned}
\vec{F}_{3\text{net}} &= -|\vec{F}_{31}|\hat{i} + |\vec{F}_{32}|\hat{i} = \left(-k\frac{q_3|q_1|}{r_{31}^2} + k\frac{q_3q_2}{r_{32}^2}\right)\hat{i} = kq_3\left(-\frac{|q_1|}{r_{31}^2} + \frac{q_2}{r_{32}^2}\right)\hat{i} \\
&= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(20 \times 10^{-6} \text{ C})\left(\frac{-80 \times 10^{-6} \text{ C}}{(0.40 \text{ m})^2} + \frac{+40 \times 10^{-6} \text{ C}}{(0.20 \text{ m})^2}\right)\hat{i} \\
&= (89.9 \text{ N})\hat{i} .
\end{aligned}$$

(b) In this case $r_{31} = 0.80 \text{ m}$ and $r_{32} = 0.60 \text{ m}$, with $\vec{F}_{31}$ directed toward $-x$ and $\vec{F}_{32}$ toward $+x$. Now we obtain

$$\begin{aligned}
\vec{F}_{3\text{net}} &= -|\vec{F}_{31}|\hat{i} + |\vec{F}_{32}|\hat{i} = \left(-k\frac{q_3|q_1|}{r_{31}^2} + k\frac{q_3q_2}{r_{32}^2}\right)\hat{i} = kq_3\left(-\frac{|q_1|}{r_{31}^2} + \frac{q_2}{r_{32}^2}\right)\hat{i} \\
&= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(20 \times 10^{-6} \text{ C})\left(\frac{-80 \times 10^{-6} \text{ C}}{(0.80 \text{ m})^2} + \frac{+40 \times 10^{-6} \text{ C}}{(0.60 \text{ m})^2}\right)\hat{i} \\
&= -(2.50 \text{ N})\hat{i} .
\end{aligned}$$

(c) Between the locations treated in parts (a) and (b), there must be one where $\vec{F}_{3\text{net}} = 0$. Writing $r_{31} = x$ and $r_{32} = x - 0.20 \text{ m}$, we equate $|\vec{F}_{31}|$ and $|\vec{F}_{32}|$, and after canceling common factors, arrive at

$$\frac{|q_1|}{x^2} = \frac{q_2}{(x - 0.20 \text{ m})^2} .$$

This can be further simplified to

$$\frac{(x - 0.20 \text{ m})^2}{x^2} = \frac{q_2}{|q_1|} = \frac{1}{2} .$$

Taking the (positive) square root and solving, we obtain $x = 0.683 \text{ m}$. If one takes the negative root and ‘solves’, one finds the location where the net force *would* be zero *if* q_1 and q_2 were of like sign (which is not the case here).

(d) From the above, we see that $y = 0$.

59. The mass of an electron is $m = 9.11 \times 10^{-31} \text{ kg}$, so the number of electrons in a collection with total mass $M = 75.0 \text{ kg}$ is

$$n = \frac{M}{m} = \frac{75.0 \text{ kg}}{9.11 \times 10^{-31} \text{ kg}} = 8.23 \times 10^{31} \text{ electrons} .$$

The total charge of the collection is

$$q = -ne = -(8.23 \times 10^{31})(1.60 \times 10^{-19} \text{ C}) = -1.32 \times 10^{13} \text{ C}.$$

60. We note that, as result of the fact that the Coulomb force is inversely proportional to r^2 , a particle of charge Q that is distance d from the origin will exert a force on some charge q_0 at the origin of equal strength as a particle of charge $4Q$ at distance $2d$ would exert on q_0 . Therefore, $q_6 = +8e$ on the $-y$ axis could be replaced with a $+2e$ closer to the origin (at half the distance); this would add to the $q_5 = +2e$ already there and produce $+4e$ below the origin, which exactly cancels the force due to $q_2 = +4e$ above the origin.

Similarly, $q_4 = +4e$ to the far right could be replaced by a $+e$ at half the distance, which would add to $q_3 = +e$ already there to produce a $+2e$ at distance d to the right of the central charge q_7 . The horizontal force due to this $+2e$ is cancelled exactly by that of $q_1 = +2e$ on the $-x$ axis, so that the net force on q_7 is zero.

61. (a) Charge $Q_1 = +80 \times 10^{-9} \text{ C}$ is on the y axis at $y = 0.003 \text{ m}$, and charge $Q_2 = +80 \times 10^{-9} \text{ C}$ is on the y axis at $y = -0.003 \text{ m}$. The force on particle 3 (which has a charge of $q = +18 \times 10^{-9} \text{ C}$) is due to the vector sum of the repulsive forces from Q_1 and Q_2 . In symbols, $\vec{F}_{31} + \vec{F}_{32} = \vec{F}_3$, where

$$|\vec{F}_{31}| = k \frac{q_3 |q_1|}{r_{31}^2}, \quad |\vec{F}_{32}| = k \frac{q_3 q_2}{r_{32}^2}.$$

Using the Pythagorean theorem, we have $r_{31} = r_{32} = 0.005 \text{ m}$. In magnitude-angle notation (particularly convenient if one uses a vector-capable calculator in polar mode), the indicated vector addition becomes

$$\vec{F}_3 = (0.518 \angle -37^\circ) + (0.518 \angle 37^\circ) = (0.829 \angle 0^\circ).$$

Therefore, the net force is $\vec{F}_3 = (0.829 \text{ N})\hat{i}$.

(b) Switching the sign of Q_2 amounts to reversing the direction of its force on q . Consequently, we have

$$\vec{F}_3 = (0.518 \angle -37^\circ) + (0.518 \angle -143^\circ) = (0.621 \angle -90^\circ).$$

Therefore, the net force is $\vec{F}_3 = -(0.621 \text{ N})\hat{j}$.

62. **THINK** We have four discrete charges in the xy -plane. We use superposition principle to calculate the net electrostatic force on particle 4 due to the other three particles.

EXPRESS Using Coulomb's law, the magnitude of the force on particle 4 by particle i is

$F_{4i} = k \frac{q_4 q_i}{r_{4i}^2}$. For example, the magnitude of $\vec{F}_{41}$ is

$$\begin{aligned} F_{41} &= k \frac{|q_4| |q_1|}{r_{41}^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(3.20 \times 10^{-19} \text{ C})(3.20 \times 10^{-19} \text{ C})}{(0.0300 \text{ m})^2} \\ &= 1.02 \times 10^{-24} \text{ N} \end{aligned}$$

Since the force is attractive, $\hat{r}_{41} = -\cos \theta_1 \hat{i} - \sin \theta_1 \hat{j} = -\cos 35^\circ \hat{i} - \sin 35^\circ \hat{j} = -0.82 \hat{i} - 0.57 \hat{j}$. In unit-vector notation, we have

$$\vec{F}_{41} = F_{41} \hat{r}_{41} = (1.02 \times 10^{-24} \text{ N})(-0.82 \hat{i} - 0.57 \hat{j}) = -(8.36 \times 10^{-25} \text{ N}) \hat{i} - (5.85 \times 10^{-24} \text{ N}) \hat{j}.$$

Similarly,

$$\begin{aligned} \vec{F}_{42} &= -k \frac{|q_4| |q_2|}{r_{42}^2} \hat{j} = -\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(3.20 \times 10^{-19} \text{ C})(3.20 \times 10^{-19} \text{ C})}{(0.0200 \text{ m})^2} \hat{j} \\ &= -(2.30 \times 10^{-24} \text{ N}) \hat{j} \end{aligned}$$

and

$$\begin{aligned} \vec{F}_{43} &= -k \frac{|q_4| |q_3|}{r_{43}^2} \hat{i} = -\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2)(6.40 \times 10^{-19} \text{ C})(3.20 \times 10^{-19} \text{ C})}{(0.0200 \text{ m})^2} \hat{i} \\ &= -(4.60 \times 10^{-24} \text{ N}) \hat{i}. \end{aligned}$$

ANALYZE (a) The net force on particle 4 is

$$\vec{F}_{4,\text{net}} = \vec{F}_{41} + \vec{F}_{42} + \vec{F}_{43} = -(5.44 \times 10^{-24} \text{ N}) \hat{i} - (2.89 \times 10^{-24} \text{ N}) \hat{j}.$$

The magnitude of the force is

$$F_{4,\text{net}} = \sqrt{(-5.44 \times 10^{-24} \text{ N})^2 + (-2.89 \times 10^{-24} \text{ N})^2} = 6.16 \times 10^{-24} \text{ N}.$$

(b) The direction of the net force is at an angle of

$$\phi = \tan^{-1} \left(\frac{F_{4y,\text{net}}}{F_{4x,\text{net}}} \right) = \tan^{-1} \left(\frac{-2.89 \times 10^{-24} \text{ N}}{-5.44 \times 10^{-24} \text{ N}} \right) = 208^\circ,$$

measured counterclockwise from the $+x$ axis.

LEARN A nonzero net force indicates that particle 4 will be accelerated in the direction of the force.

63. The magnitude of the net force on the $q = 42 \times 10^{-6}$ C charge is

$$k \frac{q_1 q}{0.28^2} + k \frac{|q_2| q}{0.44^2}$$

where $q_1 = 30 \times 10^{-9}$ C and $|q_2| = 40 \times 10^{-9}$ C. This yields 0.22 N. Using Newton's second law, we obtain

$$m = \frac{F}{a} = \frac{0.22 \text{ N}}{100 \times 10^3 \text{ m/s}^2} = 2.2 \times 10^{-6} \text{ kg}.$$

64. Let the two charges be q_1 and q_2 . Then $q_1 + q_2 = Q = 5.0 \times 10^{-5}$ C. We use Eq. 21-1:

$$1.0 \text{ N} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) q_1 q_2}{(2.0 \text{ m})^2}.$$

We substitute $q_2 = Q - q_1$ and solve for q_1 using the quadratic formula. The two roots obtained are the values of q_1 and q_2 , since it does not matter which is which. We get 1.2×10^{-5} C and 3.8×10^{-5} C. Thus, the charge on the sphere with the smaller charge is 1.2×10^{-5} C.

65. When sphere C touches sphere A , they divide up their total charge ($Q/2$ plus Q) equally between them. Thus, sphere A now has charge $3Q/4$, and the magnitude of the force of attraction between A and B becomes

$$F = k \frac{(3Q/4)(Q/4)}{d^2} = 4.68 \times 10^{-19} \text{ N}.$$

66. With $F = m_e g$, Eq. 21-1 leads to

$$y^2 = \frac{ke^2}{m_e g} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) (1.60 \times 10^{-19} \text{ C})^2}{(9.11 \times 10^{-31} \text{ kg}) (9.8 \text{ m/s}^2)}$$

which leads to $y = \pm 5.1 \text{ m}$. We choose $y = -5.1 \text{ m}$ since the second electron must be below the first one, so that the repulsive force (acting on the first) is in the direction opposite to the pull of Earth's gravity.

67. **THINK** Our system consists of two charges along a straight line. We'd like to place a third charge so that the net force on it due to charges 1 and 2 vanishes.

EXPRESS The net force on particle 3 is the vector sum of the forces due to particles 1 and 2: $\vec{F}_{3,\text{net}} = \vec{F}_{31} + \vec{F}_{32}$. In order that $\vec{F}_{3,\text{net}} = 0$, particle 3 must be on the x axis and be

attracted by one and repelled by another. As the result, it cannot be between particles 1 and 2, but instead either to the left of particle 1 or to the right of particle 2. Let q_3 be placed a distance x to the right of $q_1 = -5.00q$. Then its attraction to q_1 particle will be exactly balanced by its repulsion from $q_2 = +2.00q$:

$$F_{3x,\text{net}} = k \left[\frac{q_1 q_3}{x^2} + \frac{q_2 q_3}{(x-L)^2} \right] = k q_3 q \left[\frac{-5}{x^2} + \frac{2}{(x-L)^2} \right] = 0.$$

ANALYZE (a) Cross-multiplying and taking the square root, we obtain

$$\frac{x}{x-L} = \sqrt{\frac{5}{2}}$$

which can be rearranged to produce

$$x = \frac{L}{1 - \sqrt{2/5}} \approx 2.72 L.$$

(b) The y coordinate of particle 3 is $y = 0$.

LEARN We can use the result obtained above for consistency check. We find the force on particle 3 due to particle 1 to be

$$F_{31} = k \frac{q_1 q_3}{x^2} = k \frac{(-5.00q)(q_3)}{(2.72L)^2} = -0.675 \frac{kq q_3}{L^2}.$$

Similarly, the force on particle 3 due to particle 2 is

$$F_{32} = k \frac{q_2 q_3}{x^2} = k \frac{(+2.00q)(q_3)}{(2.72L - L)^2} = +0.675 \frac{kq q_3}{L^2}.$$

Indeed, the sum of the two forces is zero.

68. The net charge carried by John whose mass is m is roughly

$$\begin{aligned} q &= (0.0001) \frac{m N_A Z e}{M} \\ &= (0.0001) \frac{(90 \text{ kg})(6.02 \times 10^{23} \text{ molecules/mol})(18 \text{ electron proton pairs/molecule})(1.6 \times 10^{-19} \text{ C})}{0.018 \text{ kg/mol}} \\ &= 8.7 \times 10^5 \text{ C}, \end{aligned}$$

and the net charge carried by Mary is half of that. So the electrostatic force between them is estimated to be

$$F \approx k \frac{q(q/2)}{d^2} = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \frac{(8.7 \times 10^5 \text{ C})^2}{2(30 \text{ m})^2} \approx 4 \times 10^{18} \text{ N}.$$

Thus, the order of magnitude of the electrostatic force is 10^{18} N .

69. We are concerned with the charges in the nucleus (not the “orbiting” electrons, if there are any). The nucleus of Helium has 2 protons and that of thorium has 90.

(a) Equation 21-1 gives

$$F = k \frac{q^2}{r^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) (2(1.60 \times 10^{-19} \text{ C}))(90(1.60 \times 10^{-19} \text{ C}))}{(9.0 \times 10^{-15} \text{ m})^2} = 5.1 \times 10^2 \text{ N}.$$

(b) Estimating the helium nucleus mass as that of 4 protons (actually, that of 2 protons and 2 neutrons, but the neutrons have approximately the same mass), Newton’s second law leads to

$$a = \frac{F}{m} = \frac{5.1 \times 10^2 \text{ N}}{4(1.67 \times 10^{-27} \text{ kg})} = 7.7 \times 10^{28} \text{ m/s}^2.$$

70. For the net force on $q_1 = +Q$ to vanish, the x force component due to $q_2 = q$ must exactly cancel the force of attraction caused by $q_4 = -2Q$. Consequently,

$$\frac{Qq}{4\pi\epsilon_0 a^2} = \frac{Q|2Q|}{4\pi\epsilon_0 (\sqrt{2}a)^2} \cos 45^\circ = \frac{Q^2}{4\pi\epsilon_0 \sqrt{2}a^2}$$

or $q = Q/\sqrt{2}$. This implies that $q/Q = 1/\sqrt{2} = 0.707$.

71. (a) The second shell theorem states that a charged particle inside a shell with charge uniformly distributed on its surface has no net force acting on it due to the shell. Thus, inside the spherical metal shell at $r = 0.500R < R$, the net force on the electron is zero, and therefore, $a = 0$.

(b) The first shell theorem states that a charged particle outside a shell with charge uniformly distributed on its surface is attracted or repelled as if the shell’s charge were concentrated as a particle at its center. Thus, the magnitude of the Coulomb force on the electron at $r = 2.00R$ is

$$\begin{aligned} F &= k \frac{Q|e|}{r^2} = k \frac{(4\pi R^2 \sigma)|e|}{(2.0R)^2} = k\pi\sigma|e| \\ &= (8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2) \pi (6.90 \times 10^{-13} \text{ C/m}^2) (1.60 \times 10^{-19} \text{ C}) \\ &= 3.12 \times 10^{-21} \text{ N}, \end{aligned}$$

and the corresponding acceleration is

$$a = \frac{F}{m} = \frac{3.12 \times 10^{-21} \text{ N}}{9.11 \times 10^{-31} \text{ kg}} = 3.43 \times 10^9 \text{ m/s}^2.$$

72. Since the total energy is conserved,

$$\frac{1}{2} m_e v_i^2 = \frac{1}{2} m_e v_f^2 - \frac{ke^2}{r_f}$$

where r_f is the distance between the electron and the proton. For $v_f = 2v_i$, we solve for r_f and obtain

$$\begin{aligned} r_f &= \frac{2ke^2}{m_e(v_f^2 - v_i^2)} = \frac{2ke^2}{3m_e v_i^2} = \frac{2(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.6 \times 10^{-19} \text{ C})^2}{3(9.11 \times 10^{-31} \text{ kg})(3.2 \times 10^5 \text{ m/s})^2} \\ &= 1.64 \times 10^{-9} \text{ m} \end{aligned}$$

or about 1.6 nm.

73. (a) The Coulomb force between the electron and the proton provides the centripetal force that keeps the electron in circular orbit about the proton:

$$\frac{k|e|^2}{r^2} = \frac{m_e v^2}{r}$$

The smallest orbital radius is $r_1 = a_0 = 52.9 \times 10^{-12} \text{ m}$. The corresponding speed of the electron is

$$\begin{aligned} v_1 &= \sqrt{\frac{k|e|^2}{m_e r_1}} = \sqrt{\frac{k|e|^2}{m_e a_0}} = \sqrt{\frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.6 \times 10^{-19} \text{ C})^2}{(9.11 \times 10^{-31} \text{ kg})(52.9 \times 10^{-12} \text{ m})}} \\ &= 2.19 \times 10^6 \text{ m/s}. \end{aligned}$$

(b) The radius of the second smallest orbit is $r_2 = (2)^2 a_0 = 4a_0$. Thus, the speed of the electron is

$$\begin{aligned} v_2 &= \sqrt{\frac{k|e|^2}{m_e r_2}} = \sqrt{\frac{k|e|^2}{m_e (4a_0)}} = \frac{1}{2} v_1 = \frac{1}{2} (2.19 \times 10^6 \text{ m/s}) \\ &= 1.09 \times 10^6 \text{ m/s}. \end{aligned}$$

(c) Since the speed is inversely proportional to $r^{1/2}$, the speed of the electron will decrease if it moves to larger orbits.

74. Electric current i is the rate dq/dt at which charge passes a point. With $i = 0.83\text{ A}$, the time it takes for one mole of electron to pass through the lamp is

$$\Delta t = \frac{\Delta q}{i} = \frac{N_A e}{i} = \frac{(6.02 \times 10^{23})(1.6 \times 10^{-19} \text{ C})}{0.83 \text{ A}} = 1.16 \times 10^5 \text{ s} \approx 1.3 \text{ days}.$$

75. The electrical force between an electron and a positron separated by a distance r is $F_e = ke^2/r^2$. On the other hand, the gravitational force between the two charges is $F_g = Gm_e^2/r^2$. Thus, the ratio of the two forces is

$$\frac{F_e}{F_g} = \frac{ke^2/r^2}{Gm_e^2/r^2} = \frac{ke^2}{Gm_e^2} = \frac{(8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2)(1.6 \times 10^{-19} \text{ C})^2}{(6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2/\text{kg}^2)(9.11 \times 10^{-31} \text{ kg})^2} = 4.16 \times 10^{42}.$$